

Shivangi Gosain

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EDUCATION

Georgia Institute of Technology | *Bachelor of Science – Mechanical Engineering* | 2024 - May 2027 | GPA: 3.9

WORK EXPERIENCE

Boeing | May 2026 – August 2026 | *Stress Analysis Intern*

- Reviewed aircraft seat certification packages for compliance with applicable FAR regulations and Boeing engineering requirements, evaluating supplier documentation and supporting certification of business-class seat designs.
- Evaluated multiple seat load cases and load paths by reviewing FEM results at seat-leg attachments against supplier analyses; reviewed CATIA models to verify geometry and dimensions across drawings, analysis models, and certification documentation.
- Developed a Power BI dashboard that automated daily project data updates from internal software, tracking project burndown, KPIs, and delay drivers and eliminating weekly manual Excel reporting for senior engineering meetings.

GE Vernova | June 2025 – August 2025 | *Design and Analysis Engineering Intern*

- Performed lifing analysis on turbine blade coatings using ANSYS APDL, including temperature/pressure mapping, structural, creep, oxidation, and LCF simulations; validated results against field data.
- Improved component salvageability by modifying tip weld repair geometry; led feasibility study to assess increased weld depth and collaborated cross-functionally with Heat Transfer and Repair Engineering teams.
- Created a Python script to generate 3D response surface models for creep deflection prediction, improving deflection-based Operation Impact Factor (OIF) accuracy.
- Performed circumferential tolerance stack-up analysis of adjacent turbine blades, incorporating machining tolerances, coating thickness, and thermal expansion to assess interference risk under operating conditions.

Shaw Industries | June 2024 – August 2024 | *Project Engineer Intern*

- Designed a Tenter Width Monitor operating at 350°F to continuously monitor track width and detect chain deviations, supporting preventive maintenance and reducing unplanned machine downtime.
- Designed and oversaw the construction of new guarding for the shoe steamer, updated the plant layout using CAD, and assisted with the installation of a new reversing station.
- Communicated with operators and manufacturers to identify challenges and implement effective design solutions

IVue | May 2023 – August 2023 | *Engineering Intern*

- Designed a search-and-rescue drone spotlight module in Fusion 360, considering aerodynamic loads, 3D-printing material properties, and environmental operating conditions.
- Redesigned the drone cover, enhancing accessibility and modifiability of the electronics, resulting in streamlined maintenance and customization processes.

CAMPUS & COMMUNITY INVOLVEMENT

The Gift of Learning May 2020 – August 2023 *Co-Founder and Executive Director*

- Recruited 1,000+ tutors, students, and ambassadors internationally and established partnerships with nonprofits, educational institutions, and industry leaders.
- Led donation initiatives distributing 1,000+ books and \$200+ in educational supplies to underserved students in Ghana, the Philippines, and other communities.

RESEARCH & RELEVANT EXPERIENCE

Assembly Line Automation, Innovation Factory at UGA August 2023 – May 2024 *Undergraduate Researcher; Mentor: Dr. Jaime Camelio, College of Engineering*

- Researched MiR and UR10 integration in a technology-focused production facility. Utilized ROS, OpenCV, AR Tags, and various sensors to automate the process, enhancing efficiency.
- Analyzed current research to identify improvement opportunities.

TECHNICAL SKILLS

CATIA, Siemens NX, SolidWorks, Fusion 360, ANSYS, FEA, Python, MATLAB, Java, C++, ROS, OpenCV, Power BI, Manufacturing, GD&T